On tail behavior of infinite sums of independent indicators

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based on a recent joint paper with O. Iksanov (Kyiv)

Discrete Random Structures II, Będlewo, Poland

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> What is the precise asymptotic behavior of $\mathbb{P}\{Y \geq n\}$ and $\mathbb{P}\{Y = n\}$ as $n \to \infty$?

Motivation:

- range of Poissonized samples (stick around!)
- infinite Ginibre point process and decoupled renewal processes
- records in the F^α scheme

How to approach $\mathbb{P}\{Y = n\}$?
Direct approach

It turns out that $\mathbb{P}\{Y \geq n\} \sim \mathbb{P}\{Y = n\}$.

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$$\sum_{n \geq 0} \mathbb{P}\{Y = n\} z^n = \mathbb{E} z^Y = \prod_{k \geq 1} \mathbb{E} z^{\mathbb{1}_{A_k}} = \prod_{k \geq 1} (1 - r_k + r_k z)$$

How to approach $\mathbb{P}\{Y = n\}$?

Change of measure

Exponential change of measure

For each $s > 0$, introduce a new probability measure $\mathbb{P}^{(s)}$ such that

$$\mathbb{E}^{(s)}[f(Y)] = \frac{\mathbb{E}[\mathrm{e}^{sY} f(Y)]}{\mathbb{E}[\mathrm{e}^{sY}]},$$

where $f : \mathbb{R} \to \mathbb{R}$ is any measurable bounded function.

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A specialization with any $n \in \mathbb{N}$ fixed and $f(z) := \mathrm{e}^{-sz}\mathbb{1}_{\{z=n\}}$:

$$\mathbb{P}\{Y = n\} = \mathbb{E}[\mathrm{e}^{s(Y-n)}]\mathbb{P}^{(s)}\{Y = n\}$$

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Then,

$$\psi'(s) = \sum_{k\geq 1} \frac{r_k \mathrm{e}^s}{r_k \mathrm{e}^s + 1 - r_k} = \frac{\mathbb{E}[\mathrm{e}^{sY} Y]}{\mathbb{E}[\mathrm{e}^{sY}]} = \mathbb{E}^{(s)}[Y].$$

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$$\mathbb{P}\{Y = n\} = \exp(\psi(s_n) - s_n\psi'(s_n))\mathbb{P}^{(s_n)}\{Y = n\}$$

Classification
Variance

$$\mathbb{P}\{Y=n\} = \exp(\psi(s_n) - s_n\psi'(s_n))\mathbb{P}^{(s_n)}\{Y=n\} \text{ and } \mathbb{E}^{(s_n)}[Y] = n$$

$$\mathrm{Var}^{(s)} Y = \mathbb{E}^{(s)}[Y^2] - (\mathbb{E}^{(s)}[Y])^2 = \frac{\mathbb{E}[\mathrm{e}^{sY}Y^2]}{\mathbb{E}[\mathrm{e}^{sY}]} - \left(\frac{\mathbb{E}[\mathrm{e}^{sY}Y]}{\mathbb{E}[\mathrm{e}^{sY}]}\right)^2 = \psi''(s)$$

Classification
Regime 1

$$\mathbb{P}\{Y = n\} = \exp(\psi(s_n) - s_n \psi'(s_n))\mathbb{P}^{(s_n)}\{Y = n\} \text{ and } \mathbb{E}^{(s_n)}[Y] = n$$

$\psi''(s_n) \to 0$, so we expect $\mathbb{P}^{(s_n)}\{Y = n\} \to 1$

Theorem 1. As $n \to \infty$, the following are equivalent:

- (a) $\psi''(s_n) \to 0$;
- (b) $\sum_{k=1}^{n} r_k^{-1} \mathrm{e}^{-s_n} \to 0$;
- (c) $\sum_{k \geq n+1} r_k \mathrm{e}^{s_n} \to 0$.

If any of (a)-(c) holds, then

$$\mathbb{P}\{Y \geq n\} \sim \mathbb{P}\{Y = n\} \sim \exp(\psi(s_n) - s_n \psi'(s_n)), \quad n \to \infty.$$

Classification
Regime 2

$\mathbb{P}\{Y=n\} = \exp(\psi(s_n) - s_n\psi'(s_n))\mathbb{P}^{(s_n)}\{Y=n\}$ and $\mathbb{E}^{(s_n)}[Y] = n$

Theorem 2. As $n \to \infty$, the following are equivalent:

a) $\psi''(s_n) \to \infty$;
b) $\sum_{k=1}^{n} r_k^{-1} e^{-s_n} \to \infty$ and $\sum_{k \geq n+1} r_k e^{s_n} \to \infty$.

If any of (a)-(b) holds, then

$$\mathbb{P}\{Y \geq n\} \sim \mathbb{P}\{Y=n\} \sim \frac{\exp(\psi(s_n) - s_n\psi'(s_n))}{(2\pi\psi''(s_n))^{1/2}}, \quad n \to \infty.$$

Classification

Regime 3

$\mathbb{P}\{Y = n\} = \exp(\psi(s_n) - s_n\psi'(s_n))\mathbb{P}^{(s_n)}\{Y = n\}$ and $\mathbb{E}^{(s_n)}[Y] = n$

Theorem 3. Assume

(a) for each $k \in \mathbb{N}$, $r_{n+k}\mathrm{e}^{s_n} \to p_k$;

(b) for each $k \in \mathbb{N} \cup \{0\}$, $r_{n-k}^{-1}\mathrm{e}^{-s_n} \to q_k$;

(c) domination conditions.

Then

$$\mathbb{P}\{Y \geq n\} \sim \mathbb{P}\{Y = n\} \sim c_0 \exp(\psi(s_n) - s_n\psi'(s_n)), \quad n \to \infty,$$

with $c_0 := \mathbb{P}\{\sum_{m\geq 0} \theta_m = 0\} \in (0,1)$. Here, $\theta_0 \in \{-1, 0\}$ and $\theta_m \in \{-1, 0, 1\}$.

General results in action

$r_k = ck^{-\beta}$, $\beta > 1$

Theorem 4. Let $\beta > 1$ and $c > 0$. If $r_k = ck^{-\beta}$ for $k \in \mathbb{N}$, then

$$\begin{aligned}\mathbb{P}\{Y \geq n\} \;&\sim\; \mathbb{P}\{Y = n\} \\ &\sim\; \exp\Big(-\beta n \log n - \beta\Big(\log\Big(\frac{\beta\sin(\pi/\beta)}{\pi c^{1/\beta}}\Big) - 1\Big)n \\ &\quad - \frac{\beta+1}{2}\log n - \frac{\beta}{2}\log(2\beta\sin(\pi/\beta)) - \frac{1}{2}\log(2\pi) + \frac{1}{2}\log\beta\Big).\end{aligned}$$

General results in action

$r_k = c\exp(-k)$

Theorem 5. Let $c > 0$. If $r_k = c\exp(-k)$ for $k \in \mathbb{N}$, then

$$\mathbb{P}\{Y \geq n\} \sim \mathbb{P}\{Y = n\}$$
$$\sim c_0 \exp\Big(-n^2/2 + (\log c - 1/2)n - 1/8 + c_1 + h_2(1/2)\Big)$$

with known function $h_2 \sim 10^{-7}$ and known constant c_1. A constant $c_0 := \mathbb{P}\big\{\sum_{k\geq 0}\theta_k = 0\big\} \in (0,1)$.

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- $u_k = r_k(1+\varepsilon_k)$
- $\sum_{k\geq 1} |\varepsilon_k| < \infty$
- ε_k satisfy additional technical bounds

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$(\pi(t))_{t\geq 0}$ – a Poisson process of unit intensity

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independent of $\xi_1, \xi_2, \ldots$ – i.i.d. r.v. having the distribution $p_j := \mathbb{P}\{\xi_1 = j\}$ for $j \in \mathbb{N}$

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Then $\mathrm{Ran}(\xi, \pi(t)) = \sum_{k\geq 1} \mathbb{1}_{\{\pi_k(t)\geq 1\}}$ is a version of Y, with $r_k = 1 - \mathrm{e}^{-tp_k}$.

THANK YOU FOR YOUR ATTENTION!